

Element H - Prototype Testing and Data Collection Plan

Project Title: MJ's LAST Stand (Echo Rack)

H1. Addressing Requirements

- **How stable and sturdy the gong is**
- **If the gong stand is about to hold the gong**
- **How well the gong can be transported**
- **How well the mute can be implemented into the gong stand**
- **Staying on budget and completing the gong stand by the deadline**

We will test the gong stand to ensure that it works efficiently and is safe for the members of the ERHS band. These requirements were chosen when we interviewed our band teacher Mr. Waller. These are the most important things needed to effectively and efficiently create the gong stand.

H2. Conduct of Tests

We will conduct many tests to ensure our gong stand world efficiently and effectively. The tests consist of:

Durability: We will test the gong to see how strong it is when put on the stand. A gong weighs about 30 pounds so we will need to ensure that our gong stand is able to hold the gong without any problems.

Set up: We will attach the gong to the gong stand

Expected Range: The gong stand is able to hold the gong without any issues

Minimum Result: The gong stand is able to hold the gong but it struggles.

Goal: The gong is able to comfortably hold the gong.

Transportation: We will test how well the gong is able to be transported. It is very important that the gong is able to be smoothly transported because to ensure that is easy to move for concerts and events for the ERHS Band.

Set up: We will test the wheels of the gong stand on different surfaces.

Expected Range: The wheels are able to comfortably transport the gong stand.

Minimum Result: The wheels are able to work but they don't transport the gong stand smoothly.

Goal: The wheels are able to transport the gong stand effectively and efficiently.

Stability: We will test the stability of the on the gong stand to ensure that is won't fall over and will be safe to use. A gong weighs about 30 pounds and it is important the gong stand is able to remain stable while holding the gong.

Set up: We will strike the gong and see if the gong stand moves or is shaken.

Expected Range: When the gong is struck, the stand shakes minimally.

Minimum Result: When the gong is struck, the stand shakes significantly but at least does not fall over.

Goal: When the gong is struck there it remains stable and it doesn't shake.

Muting Ability: We will test the ability of the mute to dampen the sound of the gong on the gong stand. The purpose of the gong mute is to assist a percussionist in silencing the gong abruptly and swiftly after striking, without having to 'hug' the gong.

Set up: We will test the functionality of the mute by shifting the mute arms over and over again.

Expected Range: With the push of the foot pedal, the mute smoothly opens and clamps.

Minimum Result: Although it may take a lot of force or is rough, if the mute opens, its a success.

Goal: The mute opens and closes smoothly without much force.

H3. Quantity Of Tests Planned

We will do tests to address:

- How sturdy the gong stand is
- How well the mute works
- How well the gong stand is transported
- The stability of the gong.

We will do 3 trials for each test based on the requirements. This will allow to make sure the results are accurate and this will allow us to make any changes if there are any issues.

H4. Logic of the Plan

In order to successfully complete our project we will need to test different requirements to ensure our gong stand works effectively and efficiently.

We will measure:

- **Durability:** We will measure the durability of the gong stand to ensure that it works efficiently and is safe for students.
- **Mute:** We will conduct multiple tests to measure how well the mute is able to work on the gong.

- **Appearance:** We will ask students and Mr.Waller how appealing the gong looks and how we could improve it.
- **Transportation:** We will do many tests to measure how well the gong stand will be transported on different surfaces.

H5. Use of Field Experts

We have checked the testing plan with two experts in their respective fields, and they gave us helpful feedback. We talked to:

- Mr. Szczepanski (Woodworker/Professional Furniture Crafter) – He gave us advice about which woods to use and how we would go about attaching planks to one another.
- Mr. Waller (Band Teacher) – He provided us with a criteria, and a color scheme to follow to fit with the rest of the band instruments.
- Both experts provided us with valuable information and insight about what cost effective/friendly materials could be used to formulate the gong stand.

In conclusion, our testing plan makes sure the gong stand is fashionable and usable, with extreme focus on sustainability over the years. By focusing on the important design needs, doing thorough tests, and listening to experts, we are confident that our gong stand will be usable for generations of Roosevelt students, and will not diminish in safety standards.